Final Project — Part 2: Paper Reconstruction and Improvement

In this part, each group will choose one research paper related to one of the course topics:

- Score-based and diffusion models
- Compression
- Energy-based models
- GANs
- Normalizing flows
- Latent-variable models and VAEs
- Autoregressive models

The paper must include a clear model architecture (check if there is a git repository), a public dataset, and experimental results.

Stage 1 — Reconstruct the Paper

Implement the architecture described in the paper and try to reproduce its main results.

Your reconstruction should include:

- The original model architecture
- Training procedure
- Dataset or a link to the dataset if it's too big.
- Evaluation using the paper's metric (explain the meaning of the metric) and a metric studied in class

You do not need to match the paper's results exactly, but you must compare your results to the reported results to within negligible differences.

Stage 2 — Improve the Architecture

Modify the original architecture in a meaningful way.

Examples:

- Add normalization layers
- Change the encoder or decoder
- Change the latent dimension

- Add residual or attention layers
- Modify the loss function
- Change the noise schedule
- Improve the generator, discriminator, or flow coupling layer
- Add regularization
- Etc.

In a PDF file - Explain what you changed and why you expected it to improve the results. Include a table with the original results, your reconstruction results and the results of your improved paper.

Required Submission files

Submit:

1. **Code for the reconstructed model (+ the training and evaluation code)**
2. **Code for the improved model (+ the training and evaluation code)**
3. **The dataset, or a download link if the dataset is too big.**
4. **A PDF report**

PDF Report Structure:

1. **Original architecture: describe the model, loss, training objective, and important hyperparameters.**
2. **Paper results: summarize the reported results and metrics.**
3. **Reconstruction results: show your results and compare them to the paper.**
4. **Improved architecture: explain what you changed and why.**
5. **Improved results: compare (using a table) paper results, your reconstruction, and your improved model.**
6. **Discussion: explain what worked, what did not, and what you learned.**
7. **References.**